

T3 BSM UV and matched EFT contributions

EFT order: 5; loop order: 1. The EFT expressions show $\Delta\mathcal{L}_{\text{EFT}} = \mathcal{L}_{\text{matched}}^{\text{SM+BSM}} - \mathcal{L}_{\text{matched}}^{\text{SM}}$.

$$T3 - d3 - d5 - F4, \quad \alpha = 0 : \quad S_1 = (3, 0), \quad S_2 = (5, 1), \quad F = (4, \frac{1}{2})$$

$$\begin{aligned}
& \text{BSM UV Lagrangian} - \text{NewScalar1}_{j_1} \text{NewScalar1}_{j_1}^\dagger M_{S1}^2 \\
& + i \overline{N_{j_1, r_1}} \gamma^{\mu_1} D_{\mu_1} (N_{j_1, r_1}) \\
& - \overline{N_{j_1, r_1}} N_{j_1, r_1} M_F \\
& + \overline{\ell_{a_1, i_1}} C P_R \overline{N_{j_1, r_1}}^T \text{NewScalar1}_{j_1} \text{T3Y1CG}_{a_1, j_1, j_1} y_{1, i_1, r_1} \\
& + \overline{y_{2, i_1, r_1}} \overline{N_{j_1, r_1}} P_L \ell_{a_1, i_1} \text{NewScalar2}_{j_1} \overline{\text{T3Y2CG}_{a_1, j_1, j_1}} \\
& + \overline{y_{1, i_1, r_1}} \ell_{a_1, i_1}^T C P_L N_{j_1, r_1} \overline{\text{T3Y1CG}_{a_1, j_1, j_1}} \text{NewScalar1}_{j_1}^\dagger \\
& + \frac{1}{2} \text{lambdaH1Inv1} H_{a_2} \text{NewScalar1}_{j_2} \text{T3HiggsPortal1CGInv1}_{a_1, a_2, j_1, j_2} H_{a_1}^\dagger \text{NewScalar1}_{j_1}^\dagger \\
& + \frac{1}{2} \text{lambdaH1Inv1} H_{a_2} \text{NewScalar1}_{j_2} \text{T3HiggsPortal1CGInv1}_{a_1, a_2, j_2, j_1} H_{a_1}^\dagger \text{NewScalar1}_{j_1}^\dagger \\
& + \frac{1}{2} \text{lambdaH1Inv2} H_{a_2} \text{NewScalar1}_{j_2} \text{T3HiggsPortal1CGInv2}_{a_1, a_2, j_1, j_2} H_{a_1}^\dagger \text{NewScalar1}_{j_1}^\dagger \\
& + \frac{1}{2} \text{lambdaH1Inv2} H_{a_2} \text{NewScalar1}_{j_2} \text{T3HiggsPortal1CGInv2}_{a_1, a_2, j_2, j_1} H_{a_1}^\dagger \text{NewScalar1}_{j_1}^\dagger \\
& + \lambda_{\text{T3}} \text{NewScalar2}_{j_1} \overline{\text{T3MixCG}_{a_1, a_2, j_1, j_1}} H_{a_1}^\dagger H_{a_2}^\dagger \text{NewScalar1}_{j_1}^\dagger \\
& + \frac{1}{2} \text{lambdaS1Inv1} \text{NewScalar1}_{j_3} \text{NewScalar1}_{j_4} \text{T3Scalar1SelfCGInv1}_{j_1, j_3, j_2, j_4} \text{NewScalar1}_{j_1}^\dagger \text{NewScalar1}_{j_2}^\dagger \\
& + \frac{1}{2} \text{lambdaS1Inv2} \text{NewScalar1}_{j_3} \text{NewScalar1}_{j_4} \text{T3Scalar1SelfCGInv2}_{j_1, j_3, j_2, j_4} \text{NewScalar1}_{j_1}^\dagger \text{NewScalar1}_{j_2}^\dagger \\
& - M_{S2}^2 \text{NewScalar2}_{j_1} \text{NewScalar2}_{j_1}^\dagger \\
& + \overline{\ell_{a_1, i_1}} P_R N_{j_1, r_1} \text{T3Y2CG}_{a_1, j_1, j_1} y_{2, i_1, r_1} \text{NewScalar2}_{j_1}^\dagger \\
& + H_{a_1} H_{a_2} \text{NewScalar1}_{j_1} \text{T3MixCG}_{a_1, a_2, j_1, j_1} \lambda_{\text{T3}} \text{NewScalar2}_{j_1}^\dagger \\
& + \text{lambdaH2Inv1} H_{a_2} \text{NewScalar2}_{j_2} \text{T3HiggsPortal2CGInv1}_{a_1, a_2, j_1, j_2} H_{a_1}^\dagger \text{NewScalar2}_{j_1}^\dagger \\
& + \text{lambdaH2Inv2} H_{a_2} \text{NewScalar2}_{j_2} \text{T3HiggsPortal2CGInv2}_{a_1, a_2, j_1, j_2} H_{a_1}^\dagger \text{NewScalar2}_{j_1}^\dagger \\
& + \text{lambda12Inv1} \text{NewScalar1}_{j_2} \text{NewScalar2}_{j_2} \text{T3CrossScalarCGInv1}_{j_1, j_2, j_1, j_2} \text{NewScalar1}_{j_1}^\dagger \text{NewScalar2}_{j_1}^\dagger \\
& + \frac{1}{2} \text{lambdaS2Inv1} \text{NewScalar2}_{j_3} \text{NewScalar2}_{j_4} \text{T3Scalar2SelfCGInv1}_{j_1, j_3, j_2, j_4} \text{NewScalar2}_{j_1}^\dagger \text{NewScalar2}_{j_2}^\dagger \\
& + \frac{1}{2} \text{lambdaS2Inv2} \text{NewScalar2}_{j_3} \text{NewScalar2}_{j_4} \text{T3Scalar2SelfCGInv2}_{j_1, j_3, j_2, j_4} \text{NewScalar2}_{j_1}^\dagger \text{NewScalar2}_{j_2}^\dagger \\
& + \frac{1}{2} \text{lambdaS2Inv3} \text{NewScalar2}_{j_3} \text{NewScalar2}_{j_4} \text{T3Scalar2SelfCGInv3}_{j_1, j_3, j_2, j_4} \text{NewScalar2}_{j_1}^\dagger \text{NewScalar2}_{j_2}^\dagger \\
& + D_{\mu_1} (\text{NewScalar1}_{j_1})^\dagger D_{\mu_1} (\text{NewScalar1}_{j_1}) \\
& + D_{\mu_1} (\text{NewScalar2}_{j_1})^\dagger D_{\mu_1} (\text{NewScalar2}_{j_1}) \\
& \Delta \mathcal{L}_{\text{EFT}} - \frac{\sqrt{\frac{3}{2}} \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1, i_3, r_1}} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} y_{1, i_2, r_1} Y_{e, i_3, i_1} M_F^4}{2 (M_F^2 - M_{S1}^2)^2} \\
& - \frac{\sqrt{\frac{5}{2}} \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2, i_3, r_1}} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} y_{2, i_2, r_1} Y_{e, i_3, i_1} M_F^4}{2 (M_F^2 - M_{S2}^2)^2} \\
& - \frac{\sqrt{\frac{3}{2}} \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} \overline{Y_{e, i_3, i_2}} \overline{e_{i_2}} P_L \ell_{a_1, i_1} y_{1, i_3, r_1} H_{a_1}^\dagger M_F^4}{2 (M_F^2 - M_{S1}^2)^2}
\end{aligned}$$

$$\begin{aligned}
& - \frac{\sqrt{\frac{5}{2}} \hbar \log \left(\frac{\mu_{\text{bar}2}}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}{}^\dagger M_F^4}{2 (M_F^2 - M_{S_2}^2)^2} \\
& - \frac{2 \sqrt{\frac{5}{3}} \hbar \log \left(\frac{M_F^2}{M_{S_2}^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_4,i_2}}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_3} M_{S_2}^2 \lambda_{T3} \bar{e}_{a_1,a_2} \bar{e}_{a_3,a_4} M_F^3}{(M_F^2 - M_{S_1}^2) (M_F^2 - M_{S_2}^2) (M_{S_1}^2 - M_{S_2}^2)} \\
& + \frac{2 \sqrt{\frac{5}{3}} \hbar \log \left(\frac{M_F^2}{M_{S_1}^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_4,i_2}}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_3} M_{S_1}^2 \lambda_{T3} \bar{e}_{a_1,a_2} \bar{e}_{a_3,a_4} M_F^3}{(M_F^2 - M_{S_1}^2) (M_F^2 - M_{S_2}^2) (M_{S_1}^2 - M_{S_2}^2)} \\
& - \frac{2 \sqrt{\frac{5}{3}} \hbar \log \left(\frac{M_F^2}{M_{S_2}^2} \right) \overline{\lambda_{T3} \ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}^T M_{S_2}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger M_F^3}{(M_F^2 - M_{S_1}^2) (M_F^2 - M_{S_2}^2) (M_{S_1}^2 - M_{S_2}^2)} \\
& + \frac{2 \sqrt{\frac{5}{3}} \hbar \log \left(\frac{M_F^2}{M_{S_1}^2} \right) \overline{\lambda_{T3} \ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}^T M_{S_1}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger M_F^3}{(M_F^2 - M_{S_1}^2) (M_F^2 - M_{S_2}^2) (M_{S_1}^2 - M_{S_2}^2)} \\
& + \frac{\sqrt{\frac{3}{2}} \hbar \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{M_F^2 - M_{S_1}^2} \\
& + \frac{\sqrt{\frac{3}{2}} \hbar \log \left(\frac{\mu_{\text{bar}2}}{M_F^2} \right) \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{M_F^2 - M_{S_1}^2} \\
& + \frac{\sqrt{\frac{3}{2}} \hbar \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2 (M_F^2 - M_{S_1}^2)^2} \\
& + \frac{\sqrt{\frac{3}{2}} \hbar \log \left(\frac{\mu_{\text{bar}2}}{M_F^2} \right) \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{(M_F^2 - M_{S_1}^2)^2} \\
& + \frac{\sqrt{\frac{5}{2}} \hbar \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{M_F^2 - M_{S_2}^2} \\
& + \frac{\sqrt{\frac{5}{2}} \hbar \log \left(\frac{\mu_{\text{bar}2}}{M_F^2} \right) \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{M_F^2 - M_{S_2}^2} \\
& + \frac{\sqrt{\frac{5}{2}} \hbar \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2 (M_F^2 - M_{S_2}^2)^2} \\
& + \frac{\sqrt{\frac{5}{2}} \hbar \log \left(\frac{\mu_{\text{bar}2}}{M_F^2} \right) \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{(M_F^2 - M_{S_2}^2)^2} \\
& + \frac{\sqrt{\frac{3}{2}} \hbar \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}{}^\dagger M_F^2}{M_F^2 - M_{S_1}^2} \\
& + \frac{\sqrt{\frac{3}{2}} \hbar \log \left(\frac{\mu_{\text{bar}2}}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}{}^\dagger M_F^2}{M_F^2 - M_{S_1}^2} \\
& + \frac{\sqrt{\frac{3}{2}} \hbar \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} M_{S_1}^2 y_{1,i_3,r_1} H_{a_1}{}^\dagger M_F^2}{2 (M_F^2 - M_{S_1}^2)^2}
\end{aligned}$$

$$\begin{aligned}
& + \frac{\sqrt{\frac{3}{2}} \hbar \log \left(\frac{\mu \text{bar} 2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} M_{S1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{(M_F^2 - M_{S1}^2)^2} \\
& + \frac{\sqrt{\frac{5}{2}} \hbar y_{2,i_1,r_1} \overline{Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{M_F^2 - M_{S2}^2} \\
& + \frac{\sqrt{\frac{5}{2}} \hbar \log \left(\frac{\mu \text{bar} 2}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{M_F^2 - M_{S2}^2} \\
& + \frac{\sqrt{\frac{5}{2}} \hbar y_{2,i_1,r_1} \overline{Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} M_{S2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{2 (M_F^2 - M_{S2}^2)^2} \\
& + \frac{\sqrt{\frac{5}{2}} \hbar \log \left(\frac{\mu \text{bar} 2}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} M_{S2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{(M_F^2 - M_{S2}^2)^2} \\
& - \frac{2 \sqrt{\frac{5}{3}} \hbar \log \left(\frac{M_F^2}{M_{S1}^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_4,i_2}}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_3} M_{S1}^2 M_{S2}^2 \lambda_{T3} \bar{e}_{a_1,a_2} \bar{e}_{a_3,a_4} M_F}{(M_F^2 - M_{S1}^2) (M_F^2 - M_{S2}^2) (M_{S1}^2 - M_{S2}^2)} \\
& + \frac{2 \sqrt{\frac{5}{3}} \hbar \log \left(\frac{M_F^2}{M_{S2}^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_4,i_2}}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_3} M_{S1}^2 M_{S2}^2 \lambda_{T3} \bar{e}_{a_1,a_2} \bar{e}_{a_3,a_4} M_F}{(M_F^2 - M_{S1}^2) (M_F^2 - M_{S2}^2) (M_{S1}^2 - M_{S2}^2)} \\
& - \frac{2 \sqrt{\frac{5}{3}} \hbar \log \left(\frac{M_F^2}{M_{S1}^2} \right) \overline{\lambda_{T3} \ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}^T M_{S1}^2 M_{S2}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger M_F}{(M_F^2 - M_{S1}^2) (M_F^2 - M_{S2}^2) (M_{S1}^2 - M_{S2}^2)} \\
& + \frac{2 \sqrt{\frac{5}{3}} \hbar \log \left(\frac{M_F^2}{M_{S2}^2} \right) \overline{\lambda_{T3} \ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}^T M_{S1}^2 M_{S2}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger M_F}{(M_F^2 - M_{S1}^2) (M_F^2 - M_{S2}^2) (M_{S1}^2 - M_{S2}^2)} \\
& - \frac{17 \hbar B_{\mu_1 \mu_2}^2}{12 \epsilon} \\
& - \hbar \log \left(\frac{\mu \text{bar} 2}{M_F^2} \right) B_{\mu_1 \mu_2}^2 \\
& - \frac{5}{12} \hbar \log \left(\frac{\mu \text{bar} 2}{M_{S2}^2} \right) B_{\mu_1 \mu_2}^2 \\
& - \frac{6 \hbar \left(W_{\mu_1 \mu_2}^{A_1} \right)^2}{\epsilon} \\
& - 5 \hbar \log \left(\frac{\mu \text{bar} 2}{M_F^2} \right) \left(W_{\mu_1 \mu_2}^{A_1} \right)^2 \\
& - \frac{1}{6} \hbar \log \left(\frac{\mu \text{bar} 2}{M_{S1}^2} \right) \left(W_{\mu_1 \mu_2}^{A_1} \right)^2 \\
& - \frac{5}{6} \hbar \log \left(\frac{\mu \text{bar} 2}{M_{S2}^2} \right) \left(W_{\mu_1 \mu_2}^{A_1} \right)^2 \\
& + \frac{\sqrt{\frac{3}{2}} \hbar y_{1,i_3,r_1} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{2 \epsilon}
\end{aligned}$$

$$\begin{aligned}
& -\frac{1}{4}\sqrt{\frac{3}{2}}\overline{hy_{1,i_3,r_1}\ell_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1} \\
& -\frac{\sqrt{\frac{3}{2}}\overline{hy_{1,i_3,r_1}\ell_{a_1,i_2}}P_R e_{i_1}H_{a_1}M_{S1}^2y_{1,i_2,r_1}Y_{e,i_3,i_1}}{M_F^2-M_{S1}^2} \\
& -\frac{\sqrt{\frac{3}{2}}h\log\left(\frac{\mu_{S1}^{\text{bar}2}}{M_{S1}^2}\right)\overline{y_{1,i_3,r_1}\ell_{a_1,i_2}}P_R e_{i_1}H_{a_1}M_{S1}^2y_{1,i_2,r_1}Y_{e,i_3,i_1}}{M_F^2-M_{S1}^2} \\
& -\frac{\sqrt{\frac{3}{2}}\overline{hy_{1,i_3,r_1}\ell_{a_1,i_2}}P_R e_{i_1}H_{a_1}M_{S1}^4y_{1,i_2,r_1}Y_{e,i_3,i_1}}{2\left(M_F^2-M_{S1}^2\right)^2} \\
& -\frac{\sqrt{\frac{3}{2}}h\log\left(\frac{\mu_{S1}^{\text{bar}2}}{M_{S1}^2}\right)\overline{y_{1,i_3,r_1}\ell_{a_1,i_2}}P_R e_{i_1}H_{a_1}M_{S1}^4y_{1,i_2,r_1}Y_{e,i_3,i_1}}{2\left(M_F^2-M_{S1}^2\right)^2} \\
& +\frac{\sqrt{\frac{5}{2}}\overline{hy_{2,i_3,r_1}\ell_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{2\epsilon} \\
& -\frac{1}{4}\sqrt{\frac{5}{2}}\overline{hy_{2,i_3,r_1}\ell_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1} \\
& -\frac{\sqrt{\frac{5}{2}}\overline{hy_{2,i_3,r_1}\ell_{a_1,i_2}}P_R e_{i_1}H_{a_1}M_{S2}^2y_{2,i_2,r_1}Y_{e,i_3,i_1}}{M_F^2-M_{S2}^2} \\
& -\frac{\sqrt{\frac{5}{2}}h\log\left(\frac{\mu_{S2}^{\text{bar}2}}{M_{S2}^2}\right)\overline{y_{2,i_3,r_1}\ell_{a_1,i_2}}P_R e_{i_1}H_{a_1}M_{S2}^2y_{2,i_2,r_1}Y_{e,i_3,i_1}}{M_F^2-M_{S2}^2} \\
& -\frac{\sqrt{\frac{5}{2}}\overline{hy_{2,i_3,r_1}\ell_{a_1,i_2}}P_R e_{i_1}H_{a_1}M_{S2}^4y_{2,i_2,r_1}Y_{e,i_3,i_1}}{2\left(M_F^2-M_{S2}^2\right)^2} \\
& -\frac{\sqrt{\frac{5}{2}}h\log\left(\frac{\mu_{S2}^{\text{bar}2}}{M_{S2}^2}\right)\overline{y_{2,i_3,r_1}\ell_{a_1,i_2}}P_R e_{i_1}H_{a_1}M_{S2}^4y_{2,i_2,r_1}Y_{e,i_3,i_1}}{2\left(M_F^2-M_{S2}^2\right)^2} \\
& +C_{H2}H_{a_1}H_{a_1}^\dagger \\
& +\frac{\sqrt{\frac{3}{2}}\overline{hy_{1,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L\ell_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{2\epsilon} \\
& -\frac{1}{4}\sqrt{\frac{3}{2}}\overline{hy_{1,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L\ell_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger \\
& -\frac{\sqrt{\frac{3}{2}}\overline{hy_{1,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L\ell_{a_1,i_1}M_{S1}^2y_{1,i_3,r_1}H_{a_1}^\dagger}{M_F^2-M_{S1}^2} \\
& -\frac{\sqrt{\frac{3}{2}}h\log\left(\frac{\mu_{S1}^{\text{bar}2}}{M_{S1}^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L\ell_{a_1,i_1}M_{S1}^2y_{1,i_3,r_1}H_{a_1}^\dagger}{M_F^2-M_{S1}^2} \\
& -\frac{\sqrt{\frac{3}{2}}\overline{hy_{1,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L\ell_{a_1,i_1}M_{S1}^4y_{1,i_3,r_1}H_{a_1}^\dagger}{2\left(M_F^2-M_{S1}^2\right)^2} \\
& -\frac{\sqrt{\frac{3}{2}}h\log\left(\frac{\mu_{S1}^{\text{bar}2}}{M_{S1}^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L\ell_{a_1,i_1}M_{S1}^4y_{1,i_3,r_1}H_{a_1}^\dagger}{2\left(M_F^2-M_{S1}^2\right)^2}
\end{aligned}$$

$$\begin{aligned}
& + \frac{\sqrt{\frac{5}{2}} \hbar y_{2,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{2\epsilon} \\
& - \frac{1}{4} \sqrt{\frac{5}{2}} \hbar y_{2,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger \\
& - \frac{\sqrt{\frac{5}{2}} \hbar y_{2,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger}{M_F^2 - M_{S_2}^2} \\
& - \frac{\sqrt{\frac{5}{2}} \hbar \log\left(\frac{\mu_{S_2}^{\text{bar}2}}{M_{S_2}^2}\right) y_{2,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger}{M_F^2 - M_{S_2}^2} \\
& - \frac{\sqrt{\frac{5}{2}} \hbar y_{2,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_2}^4 y_{2,i_3,r_1} H_{a_1}^\dagger}{2(M_F^2 - M_{S_2}^2)^2} \\
& - \frac{\sqrt{\frac{5}{2}} \hbar \log\left(\frac{\mu_{S_2}^{\text{bar}2}}{M_{S_2}^2}\right) y_{2,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_2}^4 y_{2,i_3,r_1} H_{a_1}^\dagger}{2(M_F^2 - M_{S_2}^2)^2} \\
& - H_{a_1} \mu^2 H_{a_1}^\dagger \\
& + \frac{3 \text{lambdaH1Inv} 2^2 \hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\
& + \frac{15 \text{lambdaH2Inv} 1^2 \hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{14\epsilon} \\
& + \frac{95 \text{lambdaH2Inv} 2^2 \hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{42\epsilon} \\
& + \frac{10 \sqrt{\frac{2}{3}} \text{lambdaH2Inv} 1 \text{lambdaH2Inv} 2 \hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{7\epsilon} \\
& + \frac{3}{2} \text{lambdaH1Inv} 2^2 \hbar \log\left(\frac{\mu_{S_1}^{\text{bar}2}}{M_{S_1}^2}\right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{15}{14} \text{lambdaH2Inv} 1^2 \hbar \log\left(\frac{\mu_{S_2}^{\text{bar}2}}{M_{S_2}^2}\right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{95}{42} \text{lambdaH2Inv} 2^2 \hbar \log\left(\frac{\mu_{S_2}^{\text{bar}2}}{M_{S_2}^2}\right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{10}{7} \sqrt{\frac{2}{3}} \text{lambdaH2Inv} 1 \text{lambdaH2Inv} 2 \hbar \log\left(\frac{\mu_{S_2}^{\text{bar}2}}{M_{S_2}^2}\right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{2 \sqrt{\frac{5}{3}} \hbar \overline{\lambda_{T3}} H_{a_1} H_{a_2} \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon} \\
& + \frac{2 \sqrt{\frac{5}{3}} \hbar \overline{\lambda_{T3}} H_{a_1} H_{a_2} M_{S_1}^2 \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2 - M_{S_2}^2} \\
& + \frac{2 \sqrt{\frac{5}{3}} \hbar \log\left(\frac{\mu_{S_1}^{\text{bar}2}}{M_{S_1}^2}\right) \overline{\lambda_{T3}} H_{a_1} H_{a_2} M_{S_1}^2 \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2 - M_{S_2}^2} \\
& - \frac{2 \sqrt{\frac{5}{3}} \hbar \overline{\lambda_{T3}} H_{a_1} H_{a_2} M_{S_2}^2 \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2 - M_{S_2}^2}
\end{aligned}$$

$$\begin{aligned}
& -\frac{2\sqrt{\frac{5}{3}}\hbar\log\left(\frac{\mu_{\rm bar}^2}{M_{\rm S2}^2}\right)\overline{\lambda}_{\rm T3}H_{a_1}H_{a_2}M_{\rm S2}^2\lambda_{\rm T3}H_{a_1}^\dagger H_{a_2}^\dagger}{M_{\rm S1}^2-M_{\rm S2}^2}
\end{aligned}$$
